FIREWATCH 2026 · WSU INNOVATION WORKS × LOCKHEED MARTIN SKUNK WORKS

Firewatch

Detect → Confirm → Respond

Early wildfire warning for rural Kansas volunteer fire districts, built to work on one bar of cell.

COTTONWOOD CREEK · 450 SQ MI · 18 VOLUNTEERS



THE PROBLEM

A fire nobody sees for 31 minutes



31 min
from ignition to the first 911 call

+8 min
satellites see Kansas wildfires a median 8 minutes after someone reports them (2016–2025)

2–3 mi
of fire front when the first truck arrives at 2:52

And a system that cries wolf over March pasture burns gets switched off by April.

Detect → Confirm → Respond



Detect

One event per fire

- Co-op line faults on fire-weather days
- Satellite hotspots: VIIRS, MODIS, GOES, NGFS
- Texted reports and tower smoke sensors



Confirm

No drive-out needed

- ML score tells wildfires from burns and flares
- Two independent signals to escalate (a line fault on a red-flag day counts alone)
- Nearest neighbors texted: "See smoke? 1 yes, 2 no, 3 my burn"



Respond

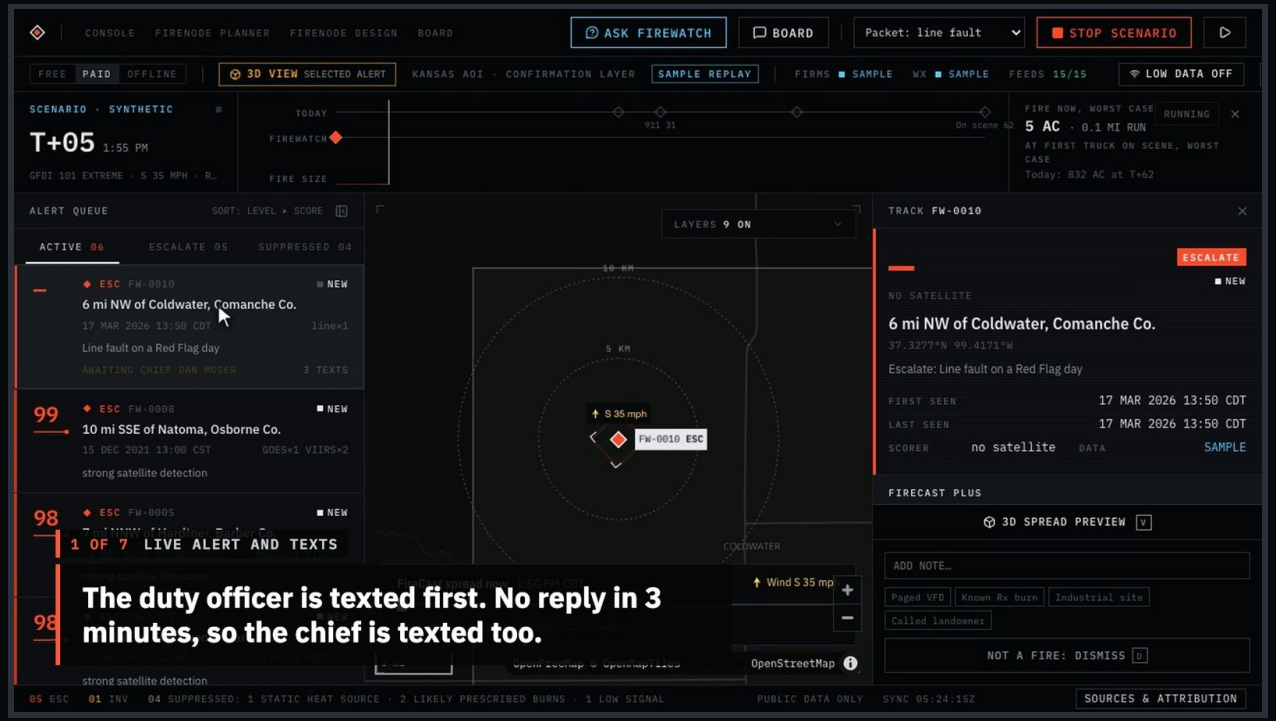
Before the first engine

- Duty officer replies PAGE by text
- Downwind ranchers get direction and arrival time
- Mutual aid drafted when crews are short

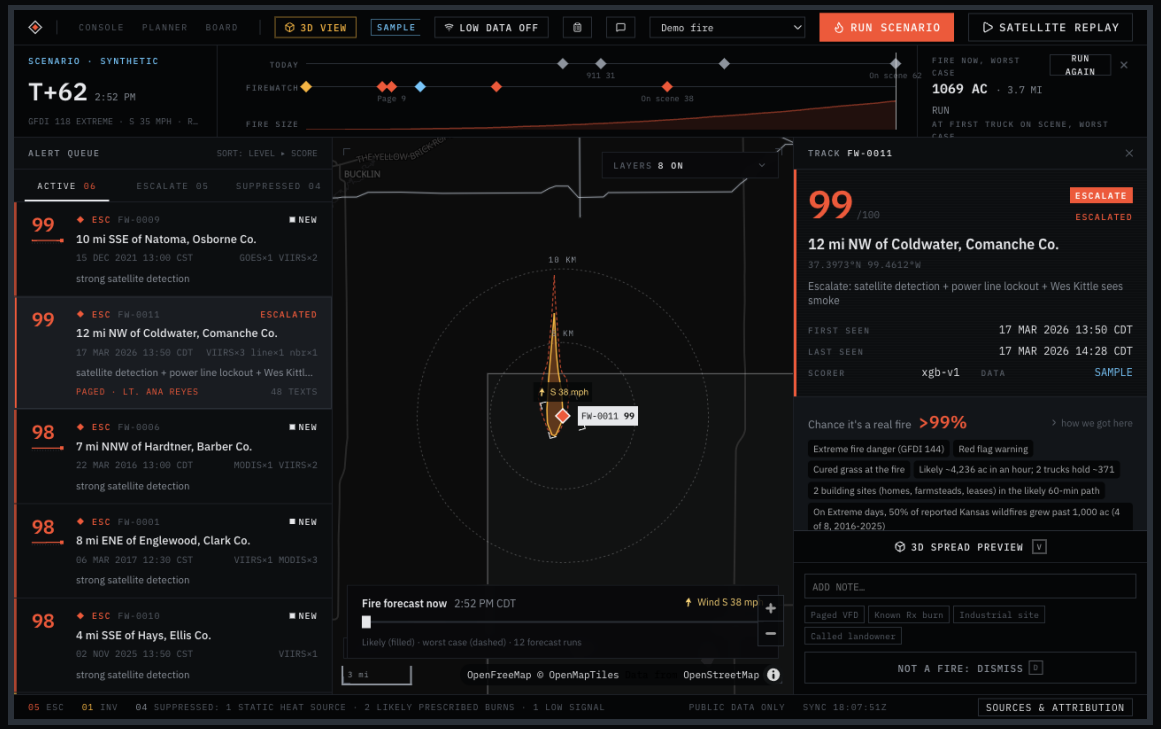
Humans decide every page. Every signal, text and decision is logged on the event.

SEE IT WORK · 1:49, NO SOUND NEEDED

The packet's fire, replayed in Firewatch



Duty officer asked at once, district paged at T+7



T+0 Co-op line locks out on a red-flag day: Escalate, the duty officer is texted

T+3 No reply in 3 minutes: the chief is texted, with a link to the alert

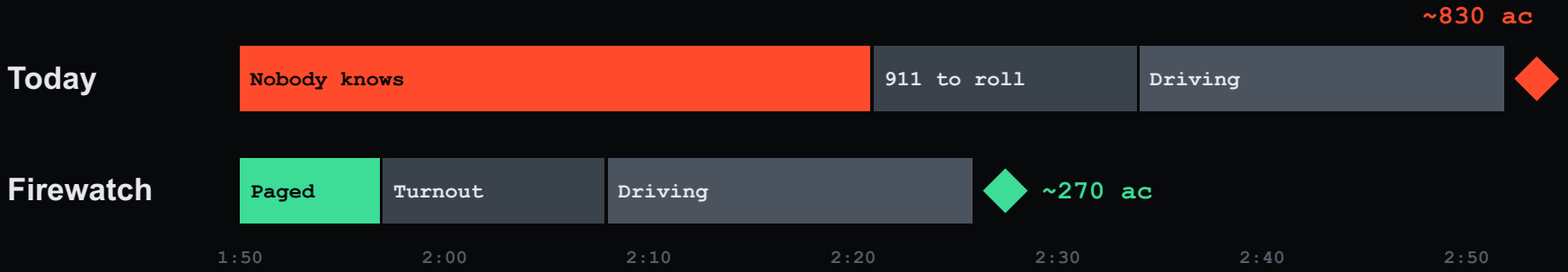
T+7 Chief replies PAGE; county dispatch tones out the district

T+7 Crew paged; neighbors with a disc or water truck asked to help

T+7 Mutual aid drafted: up to 1,030 ac in an hour, more than 2 trucks can hold

SAME FIRE, SAME CREW

First truck on scene 26 minutes sooner



T+7

district paged

today: 911 call at T+31

T+36

first truck on scene

today: T+62

~270 ac

burned when it arrives

today: ~830 acres

Fire sizes are our spread forecast's median for the packet's red-flag Tuesday, not field data. The forecast is back-tested on real Kansas fires (Anderson Creek 2016, Starbuck 2017).

NO CRYING WOLF

Quiet on burns, loud on wildfires

Refinery and plant heat kept quiet



99%

Unreported Flint Hills spring burns kept quiet



82%

Real wildfires sent to the duty officer



76%

Machine-learning scorer, tested on 2023–2025 fires it never saw

10 years

of Kansas satellite fires, matched to 224 reported wildfires and 490 burns

A person pages

Two sources to escalate, or a line fault on a red-flag day. Then someone replies PAGE.

0 texts

for a burn the landowner registered, unless it escapes

BUILT FOR ONE BAR OF CELL

Reaches people where cell doesn't

Plain text messages

Reply PAGE, 1, 2 or 3. Works on any phone, about 1 cent a text.

LoRa mesh radios

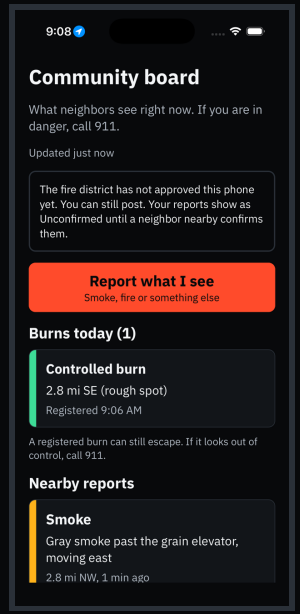
Meshtastic radios carry the same texts to trucks and ranches out of coverage. Tested with real radios.

Phone app

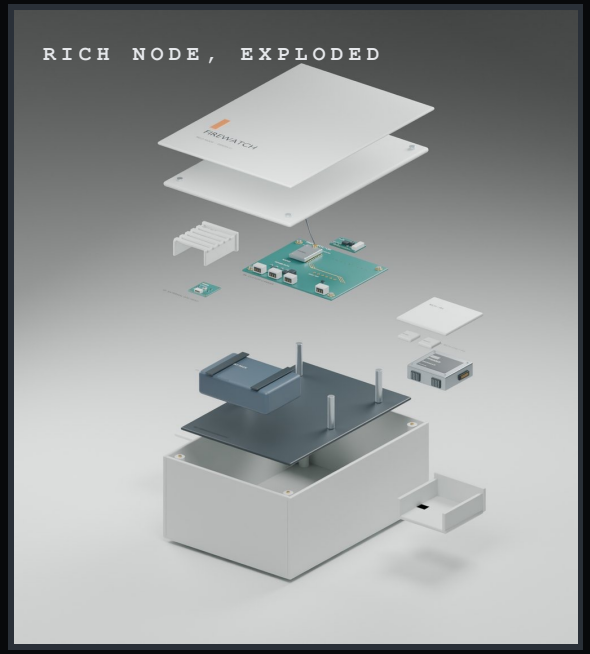
Alerts by text or radio, an offline map of where it's heading, and a board for neighbors' reports.







One box, two tiers, on a stake or a pole





Bare · ground stake



Rich · pole band

$136
bare node at ~100 units

$204
rich node, adds smoke sensing

Solar
no wiring, relays the mesh

BARE

Gas, temperature, humidity, pressure (Bosch BME688). 2 W panel, 4,400 mAh.

RICH

Adds a particle sensor that sees smoke (Plantower PMSA003I). 5 W panel, 6,600 mAh.

Blender design model, not a routed board. Detection distance, power budget and radio range are what the field burn trial measures.

Plan sensors for any Kansas county

Places 15-acre sensor plots where fires start and where the wind carries their smoke, from ten years of fires, power lines, tallgrass, terrain and hourly winds. Each plot says why it was chosen.

$15.6k — Cottonwood Creek: 10 plots, 90 nodes, 5 relays, with spares

$136 / $204 — custom-board bare and rich nodes

$0 / month — free public data, no subscriptions



WHERE IT STANDS

Built this weekend, tested on real data

BUILT AND TESTED

- ✓ 662 automated tests pass
- ✓ Scorer trained on 10 years of Kansas fires
- ✓ Spread forecast back-tested on real fires
- ✓ 13 live public data feeds checked today
- ✓ Packet scenario runs end to end, every text logged

NEXT

- → Field burn trial to measure sensor range
- → Live texting on real phones
- → Pilot with one Kansas district before spring burn season

To run a district
Free public data, about 1 cent a text, optional sensors from $136 a node



Win back the first 31 minutes.

Firewatch · Detect → Confirm → Respond

Paged at T+7 instead of T+31 · on scene at T+36 instead of T+62 · humans decide every page

`github.com/isaksmith/Firewatch`